SENTINEL EDGE MSP · ARDEPA

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TUM 2026 Security & Compliance Audit Report

Full codebase analysis · Security posture · Regulatory compliance

PROJECT	REPORT DATE	CLASSIFICATION
TUM 2026 — IT Admin Dashboard	**March 12, 2026**	**Confidential**
STACK	AUTH	DEPLOYMENT
Next.js 16 · Prisma 7 · PostgreSQL	**Auth.js v5 · Microsoft Entra ID SSO**	**Vercel (Prod) · Neon.tech (DB)**

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1 Executive Summary

Overall risk posture and key findings overview

This report presents the results of a full security, code quality, and compliance audit of the **TUM 2026 IT Admin Dashboard** — a multi-tenant MSP management platform built on Next.js 16, Auth.js v5 (Microsoft Entra ID SSO), and PostgreSQL via Prisma 7. The application manages Active Directory operations, M365 licensing, NinjaOne RMM device control, and Sophos Central endpoint security across multiple client tenants.

The audit was performed in March 2026 and covers: code cleanup, security vulnerability assessment, and regulatory compliance mapping for SOC-2, HIPAA, and CMMC Level 2.

19	13	11	0
IDOR VULNS	NO RATE LIMIT	MED FINDINGS	CONSOLE LEAKS

⚠ Critical Risk — Immediate Action Required

The most critical issue identified is a systemic **lack of tenant ownership authorization** across 19+ API routes. All routes verify that a user is authenticated, but *none* verify that the authenticated user has permission to access a specific tenant. Any authenticated user can read or modify data for any tenant by changing the `tenantId` path/query parameter. This is a classic Insecure Direct Object Reference (IDOR) vulnerability.

Key Findings Summary

Category	Count	Severity	Status
IDOR — Cross-tenant data access	19	CRITICAL	❌ Open
Missing rate limiting on device ops	13+	CRITICAL	❌ Open
Missing RBAC on mutation endpoints	6	MEDIUM	❌ Open
Error information disclosure	5	MEDIUM	❌ Open
Input validation gaps	5	MEDIUM	❌ Open
Console logging in production code	9 → 0	LOW	✅ Fixed
Schema migration (AdGroupMember)	8 files	CHANGE	✅ Deployed
M365 signInActivity 403 bug	1	HIGH	✅ Fixed
Unused imports / dead code	0	CLEAN	✅ Clean

2 Code Cleanup Summary

Items removed, refactored, and corrected during this session

Console Logging Removed (9 statements)

All `console.log`, `console.warn`, and `console.error` statements have been removed. In a multi-tenant MSP platform, log statements can expose which secrets are being accessed, which tenant operations are failing, and internal error shapes — all of which are compliance risks if logs are shipped to external aggregators.

File	Type	Was Logging
`src/lib/secrets.ts`	console.log	Vaultwarden authentication confirmation
`src/lib/secrets.ts`	console.log	Secret key name retrieval confirmation
`src/lib/secrets.ts`	console.error	Full error object on secret retrieval failure
`src/lib/secrets.ts`	console.warn	Fallback to env var with key name
`src/lib/secrets.ts`	console.log	Cache clear confirmation
`src/lib/secrets.ts`	console.log	Vaultwarden disconnect confirmation
`src/lib/audit.ts`	console.error	Audit log DB write failure
`src/lib/notify.ts`	console.error	Notification creation failure
`src/app/dashboard/technicians/actions.ts`	console.error	Technician sync error object

Bug Fixes Applied

HIGH **M365 User Detail — signInActivity causing full 403 failure**

`src/lib/graph.ts · src/app/api/tenants/[id]/users/[userId]/route.ts`

The `signInActivity` field in Microsoft Graph API requires the `AuditLog.Read.All` permission. Including it in `$select` without this permission causes the *entire* Graph call to fail with 403 — not just return null for that field. This caused the user detail endpoint to return 404 for all users on tenants without that permission.

Fix: Separated `getUserSignInActivity()` into an isolated call with internal try/catch, returning null on 403. Applied Promise.allSettled pattern to prevent cascading failures.

Schema Migration — AdGroupMember

The `tbmgroup_members_ad` table was migrated from integer FK-based design to natural string keys to match changes applied directly in Neon.tech production. This affected 8 source files:

File	Change
`prisma/schema.prisma`	Replaced groupId/userId FKs with groupSam/userSam/tenantId composite

	PK
`src/lib/ad-sync.ts`	Updated deleteMany/create to use string key pattern
`src/app/api/…/ad/groups/route.ts`	Removed member select (list endpoint)
`src/app/api/…/ad/groups/[name]/route.ts`	Two-step query: memberRecords → adUser lookup
`src/app/api/…/ad/groups/[name]/members/route.ts`	Uses userSam instead of userId
`src/app/api/…/ad/groups/[name]/members/[userSam]/route.ts`	Delete by string key
`src/app/dashboard/…/groups/page.tsx`	Replaced include with two-step query + memberCount
`src/components/ad/AdGroupMemberActions.tsx`	Flat Member interface (removed nested user object)

> ℹ️ **Note on userSam vs userUpn**
>
> The DB column is named `user_upn` for historical reasons, but it stores the SAM account name (username only, no @domain.com). The Prisma field is mapped as `userSam @map("user_upn")` to preserve the column name without an ALTER TABLE migration.

3 Security Findings

Vulnerabilities, missing controls, and exposure risks

3.1 — IDOR Vulnerabilities (Cross-Tenant Authorization)

This is the highest-priority security issue in the application. Every API route that accepts a `tenantId` (via path param `[id]` or query param) only verifies that the user is authenticated — it does *not* verify that the user is authorized to access that tenant. An attacker with any valid session can enumerate all tenants by iterating the tenant ID.

Attack Scenario

A technician at client A logs in with their Microsoft SSO. They open browser dev tools and change the tenant ID in an API call from their tenant (e.g., id=3) to another client's tenant (id=7). The API returns full Active Directory users, groups, M365 licenses, and device inventory for client 7 — without any authorization check.

Route	Risk	Severity
`/api/tenants/[id]/ad/users`	Enumerate all AD users for any tenant	CRITICAL
`/api/tenants/[id]/ad/users/[sam]` (POST)	Disable/enable/unlock/reset passwords cross-tenant	CRITICAL
`/api/tenants/[id]/ad/groups`	Enumerate all AD groups for any tenant	CRITICAL
`/api/tenants/[id]/ad/groups/[name]/members` (POST)	Add users to AD groups in any tenant	CRITICAL
`/api/tenants/[id]/ad/groups/[name]/members/[sam]` (DELETE)	Remove users from groups in any tenant	CRITICAL
`/api/tenants/[id]/ad/sync`	Trigger AD sync or read sync status for any tenant	CRITICAL
`/api/tenants/[id]/licenses`	Read M365 license data for any tenant	CRITICAL
`/api/tenants/[id]/device-assignments`	Read device assignments for any tenant	CRITICAL
`/api/cross-links` (GET + POST)	Read/create device cross-links for any tenant	CRITICAL
`/api/cross-links/auto-match`	Trigger bulk device matching for any tenant	CRITICAL
`/api/sophos/endpoints/[id]/scan`	Trigger Sophos scan on endpoints in any tenant	CRITICAL

`/api/sophos/endpoints/[id]/tamper`	Read/modify tamper protection in any tenant	CRITICAL
`/api/sophos/endpoints`	Enumerate Sophos endpoints for any tenant	CRITICAL
`/api/ninja/organizations/[orgId]`	Read NinjaOne org details without scoping	HIGH
`/api/ninja/devices`	Returns all devices across all tenants	HIGH

Recommended Fix — Tenant Authorization Middleware

```
// src/lib/tenant-auth.ts
export async function requireTenantAccess(tenantId: number, userEmail: string) {
  const session = await auth();
  if (!session?.user) throw new Error("Unauthorized");
  // ADMIN role can access all tenants
  if (session.user.role === "ADMIN") return;
  // Others: check explicit UserPermission for this tenant
  const access = await prisma.userPermission.findFirst({
    where: {
      user: { email: userEmail },
      permission: { permissionCode: `TENANT_${tenantId}` }
    }
  });
  if (!access) throw new Error("Forbidden");
}
```

3.2 — Missing Rate Limiting

No rate limiting is applied to any endpoint. The following are most critical due to their potential for abuse:

Endpoint	Risk Without Rate Limit	Severity
`/api/ninja/devices/[id]/reboot`	Spam reboot requests, disrupt endpoint operations	CRITICAL
`/api/ninja/devices/[id]/script`	Execute scripts repeatedly, amplify impact	CRITICAL
`/api/ninja/devices/[id]/services/[svc]/control`	Disable critical services in rapid succession	CRITICAL
`/api/ninja/devices/[id]/patch-apply`	Trigger patch cycles repeatedly	CRITICAL
`/api/sophos/endpoints/[id]/scan`	Resource exhaustion via endpoint scan flooding	HIGH
`/api/sophos/endpoints/[id]/tamper`	Rapidly toggle tamper protection on/off	HIGH
`/api/logs/export`	Bandwidth exhaustion via repeated large exports	MEDIUM

`/api/tenants/[id]/ad/sync`	Trigger expensive AD syncs repeatedly	MEDIUM

3.3 — Missing RBAC on Mutation Endpoints

These endpoints perform write/delete operations but only check if the user is authenticated, not whether their role allows the action:

Endpoint	Operation	Should Require
`/api/cross-links` POST	Create device cross-link	EDITOR or ADMIN
`/api/cross-links/auto-match` POST	Bulk auto-match devices	EDITOR or ADMIN
`/api/tenants/[id]/ad/sync` POST	Trigger AD full sync	EDITOR or ADMIN
`/api/tenants/[id]/ad/groups/[name]/members` POST	Add user to AD group	EDITOR or ADMIN
`/api/tenants/[id]/ad/groups/[name]/members/[sam]` DELETE	Remove user from AD group	EDITOR or ADMIN
`/api/tenants/[id]/ad/users/[sam]` POST	Disable/enable/unlock/reset password	EDITOR or ADMIN

3.4 — Input Validation Gaps

File	Parameter	Issue
`reports/task-runs/route.ts`	dateFrom, dateTo	Date strings used in `new Date()` without validation — invalid dates cause unexpected query behavior
`analytics/audit-activity/route.ts`	days	No min/max bounds — negative or huge values accepted
`ninja/devices/route.ts`	pageSize	No upper bound — DoS potential with very large values
`logs/export/route.ts`	filterFrom, filterTo	Date strings parsed without validation
`custom-fields/values/route.ts`	entityId	Stored without entity existence check — orphaned values possible

3.5 — Error Information Disclosure

Several routes return raw error messages from third-party APIs (NinjaOne, Sophos) directly to the client. These messages can reveal internal system details, API version information, rate limit state, or credential issues.

File	Disclosed Information

`ninja/devices/route.ts`	Raw NinjaOne error message + HTTP status
`sophos/endpoints/[id]/scan/route.ts`	Raw Sophos API error response
`sophos/endpoints/route.ts`	Raw Sophos API error response
`cross-links/auto-match/route.ts`	Raw internal error message

Recommendation: Return generic error messages to clients (`{ error: "Operation failed" }`), log full error details server-side only.

4 SOC-2 Compliance Analysis

AICPA Trust Services Criteria — Type II readiness assessment

SOC-2 Type II evaluates an organization's controls over security, availability, processing integrity, confidentiality, and privacy over a period of time. As an MSP platform with access to client Active Directory, endpoint security, and M365 environments, TUM 2026 has significant SOC-2 obligations.

CC6 — Logical and Physical Access Controls

Criteria	Description	Status
CC6.1	Logical access security measures	PARTIAL — Auth present, no tenant isolation
CC6.2	Prior to issuance of access credentials, registration and authorization	PARTIAL — Role model exists, RBAC gaps on mutations
CC6.3	Removes access when no longer required	PARTIAL — SSO session expiry works; no explicit revocation flow
CC6.6	Restricts access to information assets authorized	FAIL — IDOR vulnerabilities allow cross-tenant access
CC6.7	Restricts transmission and removal of information	PARTIAL — HTTPS enforced; export endpoint lacks rate limit
CC6.8	Prevents or detects unauthorized/malicious software	N/A — Platform-level control (Vercel/Neon handle this)

CC7 — System Operations

Criteria	Description	Status
CC7.1	Monitors configuration of infrastructure	PARTIAL — Audit log exists; no automated alerting
CC7.2	Monitors system components for anomalies	FAIL — No anomaly detection; no alerting on auth failures
CC7.3	Evaluates security events	FAIL — Audit log captured but not reviewed/alerted on
CC7.4	Responds to identified security events	FAIL — No incident response procedure documented

CC8 — Change Management

Criteria	Description	Status
CC8.1	Authorizes, designs, develops, configures changes	PARTIAL — Git history exists; no formal PR review policy

CC9 — Risk Mitigation

Criteria	Description	Status
CC9.1	Identifies, selects and develops risk mitigation	FAIL — No formal risk register maintained
CC9.2	Assesses and manages vendor risk	PARTIAL — Vercel, Neon, Microsoft, NinjaOne, Sophos used without formal BAAs

Confidentiality & Privacy

- ✓ C1.1 — Identifies confidential information (AD user data, M365 data marked as tenant-confidential by design)
- ✗ C1.2 — Disposes confidential information — no data retention/deletion policy for AD sync data
- ✗ P1.1 — Privacy notice — no privacy policy accessible to end users
- ✗ P4.1 — Collects PII with consent — no consent mechanism for syncing employee data

SOC-2 Readiness Score: ~35%

To achieve SOC-2 Type II certification, the following must be addressed: (1) Fix all IDOR vulnerabilities to satisfy CC6.6, (2) Implement automated alerting on audit events for CC7.2-7.4, (3) Document an incident response policy for CC7.4, (4) Add data retention/disposal controls for C1.2, (5) Execute BAAs with all data processors (Vercel, Neon.tech).

5 HIPAA Compliance Analysis

45 CFR Parts 160 and 164 — Security Rule Assessment

While TUM 2026 is primarily an IT operations tool (not a healthcare application), MSP clients often include healthcare organizations. If the platform is used to manage IT infrastructure for entities covered under HIPAA (hospitals, clinics, insurance companies), the MSP operating TUM 2026 may qualify as a **Business Associate** and must comply with HIPAA's Security Rule for any PHI that could flow through the system (e.g., AD user accounts for healthcare staff).

Administrative Safeguards (§ 164.308)

Safeguard	Requirement	Status
Security Officer	Designated security official	GAP — Not documented in system
Workforce Training	Security awareness training for all staff	GAP — Not tracked in the platform
Access Management	Formal process for access granting/revocation	PARTIAL — Role system exists; no formal access review workflow
Contingency Plan	Data backup, disaster recovery, emergency mode	PARTIAL — Neon.tech provides automated backups; no DR procedure documented
Audit Controls	Hardware, software, and procedural mechanisms for activity review	PARTIAL — Audit log table exists; not reviewed, alerted on, or retained per policy
BA Agreements	Business Associate Agreements with vendors	FAIL — No BAAs with Vercel, Neon.tech, Microsoft, NinjaOne, Sophos documented

Technical Safeguards (§ 164.312)

Safeguard	Requirement	Status
Access Control	Unique user identification, automatic logoff, encryption/decryption	PARTIAL — SSO unique IDs ✓; no automatic logoff configured; IDOR allows cross-access ✗
Audit Controls	Record and examine activity in PHI information systems	PARTIAL — Audit log present; no automated review
Integrity	Authenticate or corroborate PHI has not been improperly altered	PARTIAL — DB constraints + Prisma type safety; no cryptographic integrity verification
Transmission Security	Encrypt PHI transmitted over open networks	PASS — HTTPS enforced via Vercel + Neon TLS
Unique User ID	Assign unique name/number for identifying/tracking user identity	PASS — Microsoft SSO provides unique user IDs

Encryption at Rest	Encrypt PHI at rest (addressable)	PARTIAL — Neon.tech provides encryption at rest; AD passwords never stored in app ✓

HIPAA Readiness Score: ~50%

If TUM 2026 manages any HIPAA-covered tenants, the highest priority gaps are: (1) Execute BAAs with all vendors, (2) Fix IDOR vulnerabilities to meet Access Control requirements, (3) Implement audit log review + retention policy (minimum 6 years under HIPAA), (4) Document a contingency plan with RTO/RPO targets, (5) Configure automatic session expiration.

6 CMMC Level 2 Compliance Analysis

Cybersecurity Maturity Model Certification — 110 NIST SP 800-171 Practices

CMMC Level 2 requires 110 practices aligned with NIST SP 800-171, focused on protecting Controlled Unclassified Information (CUI). If the MSP's clients include defense contractors or federal agencies, TUM 2026 may be used to manage infrastructure that handles CUI — creating CMMC obligations for the MSP itself.

Access Control (AC) Domain

- ✓ AC.1.001 — Limit access to authorized users (SSO enforces authentication)
- ✗ AC.1.002 — Limit access to authorized transactions — IDOR allows cross-tenant operations
- ✓ AC.2.005 — Provide privacy and security notices — login page exists (partial)
- ✗ AC.2.006 — Limit use of portable storage — no controls; export endpoint unrestricted
- ✗ AC.2.007 — Employ the principle of least privilege — VIEWER role users can trigger mutations on some endpoints
- ✗ AC.2.009 — Limit unsuccessful logon attempts — no lockout policy; no brute-force protection
- ✗ AC.2.013 — Monitor and control remote access sessions — no session monitoring or anomaly detection
- ✗ AC.3.017 — Separate duties of individuals to reduce risk — single ADMIN role has unlimited access; no separation of duties
- ✗ AC.3.018 — Prevent non-privileged users from executing privileged functions — RBAC gaps on mutations allow this

Audit and Accountability (AU) Domain

- ✓ AU.2.041 — Ensure actions of individual users can be traced (AuditLog model captures actor, action, entity)
- ✗ AU.2.042 — Create and retain system audit logs — no defined retention period; logs never purged but also never explicitly retained
- ✗ AU.3.045 — Review and update logged events — no log review process or automated alerting
- ✗ AU.3.046 — Alert in the event of audit process failure — no alerting on audit failures

Configuration Management (CM) Domain

- ✓ CM.2.061 — Establish baselines for system configuration (package.json + Prisma schema define baselines)
- ⚠ CM.2.062 — Establish and enforce security configuration settings — .env.local not in repo ✓; no hardened config policy
- ✗ CM.3.068 — Restrict, disable, or prevent use of nonessential programs — no endpoint for disabling specific API routes

Incident Response (IR) Domain

- ✗ IR.2.092 — Establish operational incident-handling capability — no IR plan
- ✗ IR.2.093 — Track, document, and report incidents — no incident tracking in the application
- ✗ IR.3.098 — Test incident response capability — no documented testing

System and Communications Protection (SC) Domain

- ✓ SC.1.175 — Monitor, control, and protect communications at external boundaries (HTTPS via Vercel)
- ✓ SC.3.177 — Employ FIPS-validated cryptography — TLS 1.3 on Vercel + Neon
- ✗ SC.3.187 — Establish and manage cryptographic keys — Vaultwarden integration present but auth keys stored as env vars in Vercel dashboard (not HSM-managed)

CMMC Level 2 Readiness Score: ~30%

The most critical gap is AC domain failures — specifically IDOR vulnerabilities violating AC.1.002 and AC.2.007 (least privilege). CMMC requires a formal System Security Plan (SSP) documenting all 110 practices; this must be created and maintained.

7 Improvement Roadmap

Recommended improvements from three expert perspectives

7.1 — IT Security Specialist Perspective

Phase 1 — Immediate (0–2 weeks)

CRITICAL Implement tenant ownership authorization middleware and apply to all /tenants/[id]/* routes. Create a `requireTenantAccess(tenantId)` helper in `src/lib/tenant-auth.ts` and add ADMIN bypass (ADMINs can access all tenants).

CRITICAL Add `requireRole("EDITOR")` checks to all mutation endpoints that currently lack RBAC (cross-links POST, AD sync POST, group member add/remove, AD user operations).

CRITICAL Implement rate limiting using `next-rate-limit` or Upstash Redis: 10 device actions/hour/user for reboot/script/patch; 5 scans/hour/user for Sophos; 3 exports/hour/user for logs.

HIGH Replace raw third-party API error messages with generic client-facing errors. Log full details server-side only.

Phase 2 — Short Term (2–6 weeks)

HIGH Add input validation using `zod` on all API route handlers: validate date ranges, numeric bounds, and allowed enum values before DB queries.

HIGH Implement automated security alerting: notify ADMIN users via the notification system when audit logs detect suspicious patterns (multiple failed AD sync attempts, cross-tenant access attempts, excessive device operations).

HIGH Configure automatic session expiration: set JWT `maxAge` in NextAuth config (recommend 8 hours for production MSP use).

MEDIUM Add Content Security Policy (CSP) headers to prevent XSS. Vercel `next.config.mjs` supports custom security headers.

Phase 3 — Medium Term (1–3 months)

MEDIUM Implement MFA enforcement: require all users to have MFA enabled on their Microsoft account before allowing SSO login (Graph API can check MFA registration status).

MEDIUM Create a security dashboard showing: failed login attempts, cross-tenant access patterns, device action frequency, audit log anomalies. Expose this to ADMINs only.

MEDIUM Implement audit log retention policy and export-to-SIEM capability (Splunk, Sentinel, Elastic) for compliance.

7.2 — MSP Specialist Perspective

Client Data Isolation & Trust

CRITICAL Implement the tenant isolation model: each client should only see their own data. Consider a `tenantUsers` junction table linking dashboard users to authorized tenants. This is the foundational trust requirement for any MSP platform.

HIGH Add per-tenant permission granularity: instead of global ADMIN/EDITOR/VIEWER, support tenant-scoped roles (e.g., Tenant A VIEWER, Tenant B EDITOR). This allows client self-service without cross-tenant risk.

HIGH Add a client portal mode: read-only view where clients can log in with their own Microsoft accounts and see only their tenant data, without exposing other clients or MSP-internal data.

Operational Excellence

HIGH Add scheduled AD sync: implement a cron job (Vercel cron or external scheduler) to auto-sync AD data every 4–8 hours, replacing the current manual sync requirement.

MEDIUM Add a tenant health dashboard: aggregate key metrics per client (AD user count, disabled accounts %, locked accounts, expiring passwords, Sophos coverage %, NinjaOne offline devices) into a single executive view.

MEDIUM Implement ticket system integration: when performing AD operations (password reset, account enable/disable), optionally auto-create/close tickets in ConnectWise Manage or Autotask for documentation and billing.

MEDIUM Add SLA tracking: for each task run, track time-to-resolution and flag SLA breaches. Report on technician response times per client.

7.3 — Senior Developer Perspective

Code Architecture

HIGH Extract a `withTenantAuth(tenantId, handler)` higher-order function that handles authentication + tenant ownership check. Apply it to all 19 vulnerable routes. This eliminates ~150 lines of duplicated auth boilerplate and ensures consistent behavior.

HIGH Add Zod schema validation at API boundaries. Create a `src/lib/schemas/` directory with reusable validators. This makes input validation composable and testable.

MEDIUM Eliminate the N+1 query pattern in AD group member lookup: the current two-step query (memberRecords → adUser findMany) is correct but could be optimized with a Prisma `JOIN` equivalent via `include` once the schema supports it.

MEDIUM Add end-to-end TypeScript types for all API responses. Currently, client-side components use implicit types. Add a `src/types/api.ts` file with typed response shapes for all major endpoints.

Testing & Reliability

HIGH Add integration tests for IDOR protection: once tenant authorization is implemented, add test cases that verify cross-tenant access is rejected with 403. This ensures regressions are caught in CI.

MEDIUM Add a health check endpoint (`/api/health`) that verifies DB connectivity, Vaultwarden accessibility, and Graph API token freshness. Use this in Vercel's uptime monitoring.

MEDIUM Implement structured logging with a library like `pino` or `winston`. Replace the removed `console.*` calls with structured log events that include tenant context, user ID, and request tracing for production observability without sensitive data leakage.

MEDIUM Add Prisma query timeout configuration to prevent long-running queries from blocking serverless function execution time limits on Vercel.

8 Appendix

Technical reference, environment, and methodology

Application Architecture Overview

Layer	Technology	Purpose
Frontend	Next.js 16 App Router + Tailwind CSS v4	Server-rendered UI with React Server Components
Authentication	Auth.js v5 (NextAuth) + Microsoft Entra ID	SSO via OAuth2/OIDC, JWT session strategy
API Layer	Next.js Route Handlers	REST API for all data operations
ORM	Prisma 7 + prisma.config.ts	Type-safe database access with migrations
Database	PostgreSQL (Neon.tech prod, localhost dev)	Serverless PostgreSQL with branching
Secrets	Vaultwarden (Bitwarden SDK)	Self-hosted secret management with in-memory caching
AD Integration	SentinelAgent + PowerShell	On-premise AD operations via agent bridge
M365	Microsoft Graph API (client credentials)	Per-tenant App Registration, users/groups/licenses
RMM	NinjaOne API (OAuth2)	Device inventory, maintenance, patching, scripts
Security	Sophos Central Partner API	Endpoint protection, scan, tamper protection
Deployment	Vercel (serverless)	CI/CD via GitHub Actions
Notifications	SSE + in-app notification center	Real-time user notifications

Audit Methodology

This audit was performed through static code analysis of the full TUM 2026 source tree. All API route handlers (`route.ts` files), library modules (`src/lib/`), and middleware were reviewed. The following checks were performed:

- ✓ Authentication verification (session?.user checks)
- ✓ Authorization verification (tenant ownership cross-checks)
- ✓ RBAC enforcement (requireRole() calls)
- ✓ Console logging statements
- ✓ Input validation and sanitization
- ✓ Error handling and information disclosure
- ✓ Hardcoded credentials and configuration

- ✓ Rate limiting controls
- ✓ Unused imports and dead code

Compliance Standards Referenced

Standard	Version / Reference	Scope
SOC-2	AICPA 2017 Trust Services Criteria	Security, Availability, Confidentiality
HIPAA	45 CFR Parts 160 and 164 (Security Rule)	PHI in MSP client environments
CMMC	Level 2 (NIST SP 800-171 Rev 2, 110 practices)	CUI in defense contractor environments
OWASP	OWASP Top 10 2021	A01 Broken Access Control (IDOR), A09 Logging Failures

Document Information

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Next Review	June 2026 (or after major feature releases)

Disclaimer

This report is based on static code analysis only. It does not include dynamic testing (penetration testing), runtime behavior analysis, or infrastructure security assessment. A full security audit would require additional dynamic testing, infrastructure review, and manual penetration testing by a certified security professional.